

# PRASHANT GAUTAM

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Licensed Civil engineer and Pulchowk Campus alumnus with hands-on expertise in Structural Resilience, Post-fire forensic investigation, and non-destructive material testing. Experienced in correlating field diagnostic data with structural theory to evaluate infrastructural compromise. Seeking to leverage background in structural engineering and forensic reporting to pursue a M.S. graduate program focused on post fire assessment and transportation infrastructure resilience.

## EDUCATION

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Pulchowk Campus, Institute of Engineering (IOE), Tribhuvan University (T.U)

2025

### *Bachelor's in Civil Engineering*

*Awarded with regular program scholarship provided by Institute of Engineering (IOE).*

## RESEARCH INTEREST

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- **Transportation Infrastructure Resilience:** Post-Fire detailed damage assessment and structural health monitoring of tunnels, highway bridges, and critical transit corridors.
- **Forensic Engineering and Non-Destructive testing (NDT):** Utilizing advanced diagnostics (UPV, Rebound Hammer, Core sampling, etc.) to evaluate thermal-induced degradation and residual capacity of structural materials.
- **Infrastructure Rehabilitation and Asset Management:** Developing predictive data-driven framework to guide "Repair vs. Demolition" outcomes for public infrastructure subjected to extreme environmental hazards.

## RELEVANT EXPERIENCE

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Institute of Engineering Consultancy Service (IOECS), Pulchowk, Lalitpur  
*Civil Engineer (Structural Assessment)*

Oct 2025 – June 2026

- **Post-Fire Forensic Investigation:** Contributed to Post-fire Detailed Damage Assessments (DDA) for 44 administrative and residential structures of federal and central government in Kathmandu and Pokhara to evaluate fire-induced structural compromise and community safety.
- **Material Testing (SDT & NDT):** Executed Concrete Core Cutting, Carbonation Depth Analysis, and NDTs (Rebound Hammer, UPV) to determine residual material strength and durability.
- **Forensic Reporting:** Correlated field data with structural observations to assist in drafting forensic reports determining 'Repair vs. Demolition' outcomes.
- **Estimation & BOQ:** Developed comprehensive Cost Estimates and Bill of Quantities (BOQ) for rehabilitation works based on Government of Nepal norms and current district rates.

## CORE ACADEMIC PROJECTS

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Undergraduate Project: Seismic Resistant Design of Multi-Storied RCC Commercial Building

2024 - 2025

- Performed comprehensive structural modeling and dynamic analysis of a B+G+9 storied building using **ETABS** to ensure structural stability and seismic resilience.
- Designed structural members (Beams, Columns, Footings) in compliance with **IS 1893:2016** and **IS 13920** (Ductile Detailing) to optimize performance under earthquake loads.
- Conducted **Equivalent Static** and **Response Spectrum Analysis** to verify drift limits and structural stability as per the Nepal National Building Code (**NBC**).

## INDUSTRIAL ENGINEERING EXPERIENCE

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**Divyashwori Nirman Sewa, Banepa, Kavrepalanchowk**  
*Civil Engineer (Site Supervision and Quality Assurance)*

**July 2025 – Present**

- **Structural Code Adherence:** Supervised daily site construction activities to guarantee strict compliance with approved structural engineering drawings, design standards, and NBC codes.
- **Material Quality Controls:** Led reinforcement inspections (Bar Bending Schedules - BBS) and evaluated formwork stability alongside routine concrete workability/slump tests prior to structural pours.
- **Technical Measurements:** Conducted precise on-site quantity take-offs to verify structural dimensions and maintain high transparency in engineering data logs.

**D.S. Nirman Sewa, Chauri Deurali, Kavrepalanchowk**  
*Civil Engineer Intern*

**Sept 2024 – June 2025**

- **Independent Site Layouts:** Took independent charge of translating blueprint dimensions onto the physical site, handling layouts and basic surveys.
- **Material Tracking:** Oversaw reinforcement steel inspections and handled everyday quantity take-offs to ensure materials were used efficiently within budget.

## SKILLS AND TOOLS

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### Engineering Software:

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|-----------|----------------------------|
| • E-Tabs  | • SW-DTM                   |
| • AutoCAD | • Advanced Microsoft Excel |

### Diagnostic Equipment (NDT):

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|-----------------------------------------|------------------------------------|
| • Ultrasonic Pulse Velocity (UPV) Meter | • Phenolphthalein Carbonation Test |
| • Rebound Hammer                        | • Core Cutting                     |

### Survey Instruments:

- |                 |               |
|-----------------|---------------|
| • Total Station | • Auto Level, |
| • Theodolite    | • Compass     |

### Core Competencies:

- |                                |                            |
|--------------------------------|----------------------------|
| • Structural Diagnostics       | • Quantity Estimation      |
| • Post-Fire Forensic Reporting | • Concrete Quality Control |
| • Seismic Design               |                            |

## PROFESSIONAL DEVELOPMENT & LEADERSHIP

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- **Technical Certification:** Completed an ADB-sponsored capacity-building course on Road Slope Stability and Bioengineering (RCIP-AF TA-10113 NEP).
- **Team Leadership:** Stepped as a project group team leader and served as an Academic Class Representative
- **Magazine Editor:** Worked as the coordinator and editor for “The Bridge” Magazine Published by CESS-Nepal
- **Exhibition Management:** Led the student organizing committee as the primary coordinator to plan, fundraise, and execute a campus-wide health camp sponsored by CESS-Nepal; volunteered as a core technical coordinator for "AADHAR-2081", the premier Civil Engineering Exhibition at Pulchowk Campus.